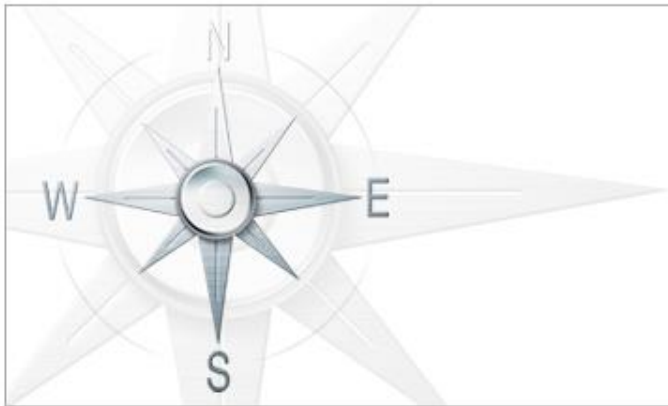




สำนักงานส่งเสริมอุตสาหกรรมซอฟต์แวร์แห่งชาติ (องค์การมหาชน)
Software Industry Promotion Agency (Public Organization)

The Development of the Thai National Grid Center

Dr.Tiranee Achalakul
Representative,
Thai National Grid Center
Software Industry Promotion Agency (SIPA)
Ministry of Information and Communication
Technology





Motivation

- Today Thai researchers are developing new and innovative ideas but the lack of proper computational infrastructure are slowing down the process.
- How can the computing power be provided sufficiently for large scale simulations?
- The solution is an introduction of a new computing infrastructure that can aid the sustainable research and development.



The Tri-C E-Infrastructure

Connectivity

- Powerful network infrastructure that link all resources together

Computation

- Large and powerful computing resources
- Large and scalable data storage

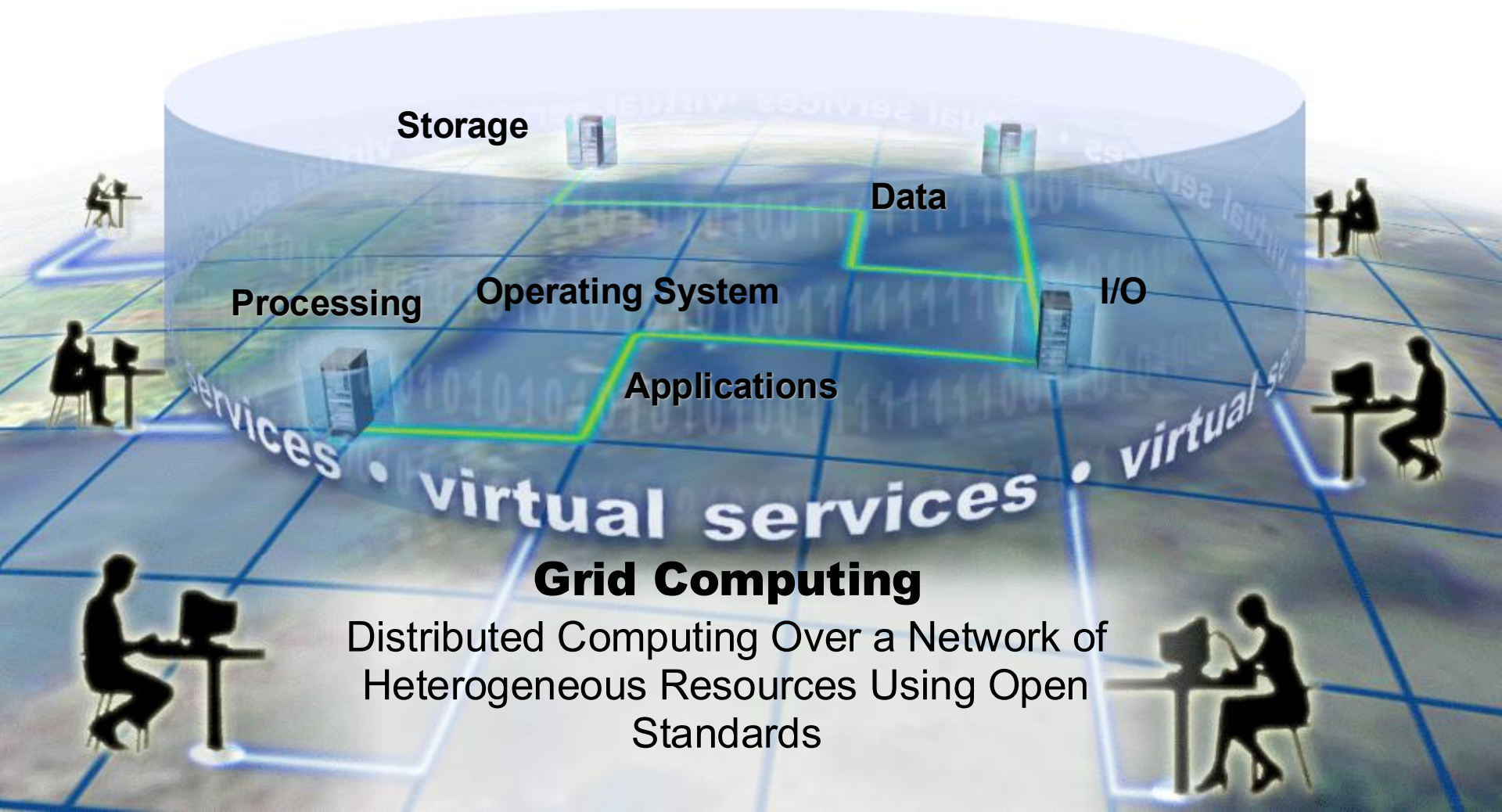
Convenience

- Simple and easy-to-use environment for users
- Powerful and rich set of services



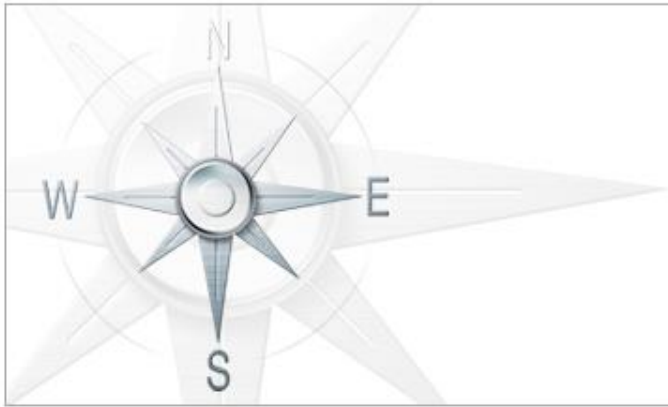
ThaiGrid
Thai National Grid Project

Grid Computing: The solution

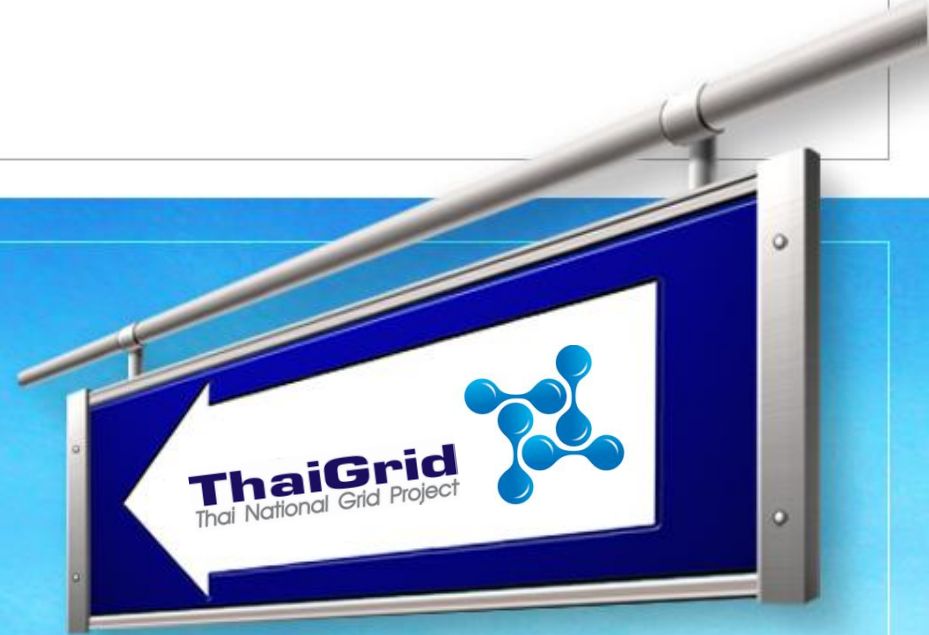


Grid Computing

Distributed Computing Over a Network of
Heterogeneous Resources Using Open
Standards



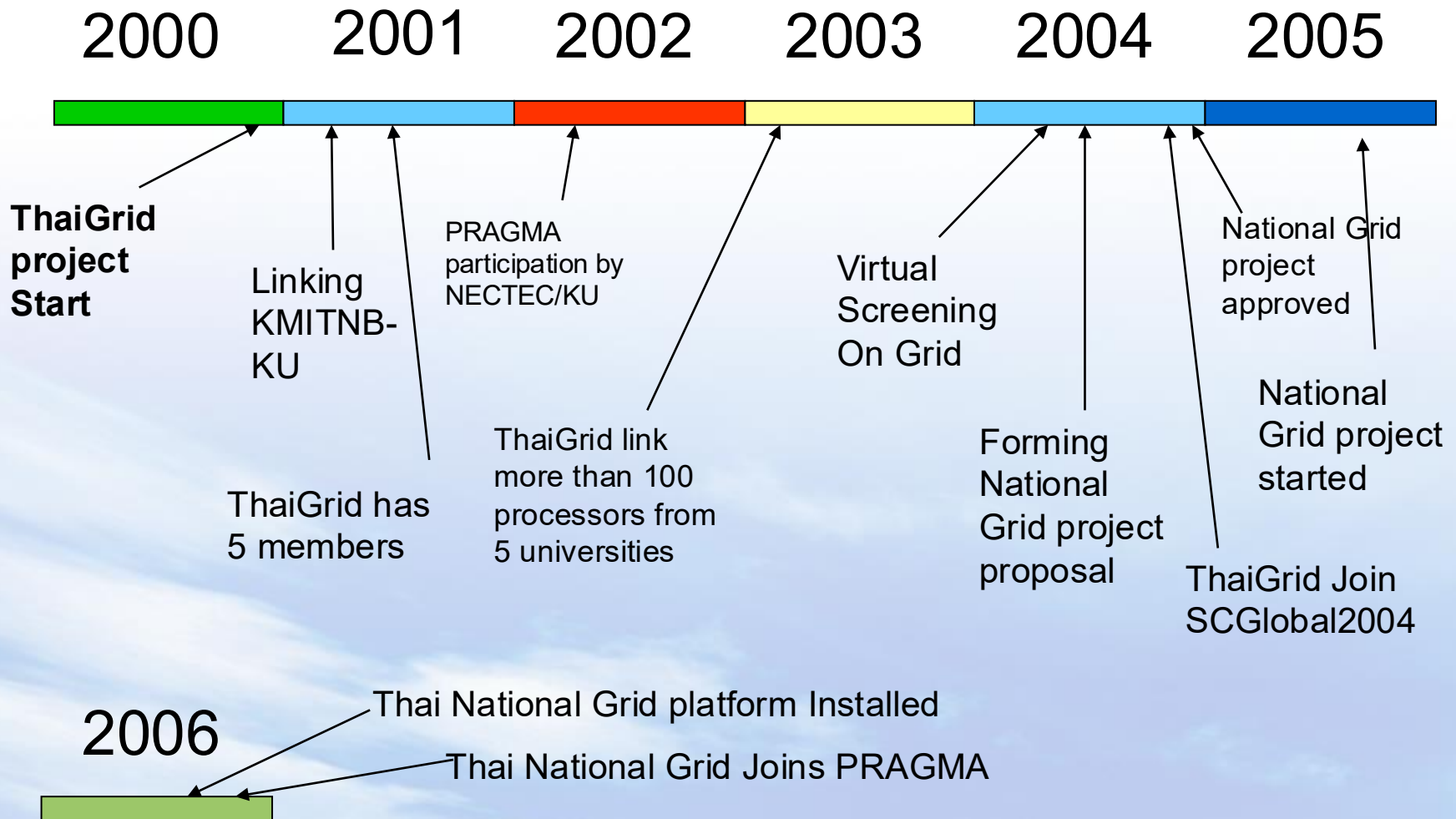
สำนักงานส่งเสริมอุตสาหกรรมซอฟต์แวร์แห่งชาติ (องค์การมหาชน)
Software Industry Promotion Agency (Public Organization)



The Efforts at TNGC



ThaiGrid Development Efforts from 2000 to 2006



Thai National Grid Project



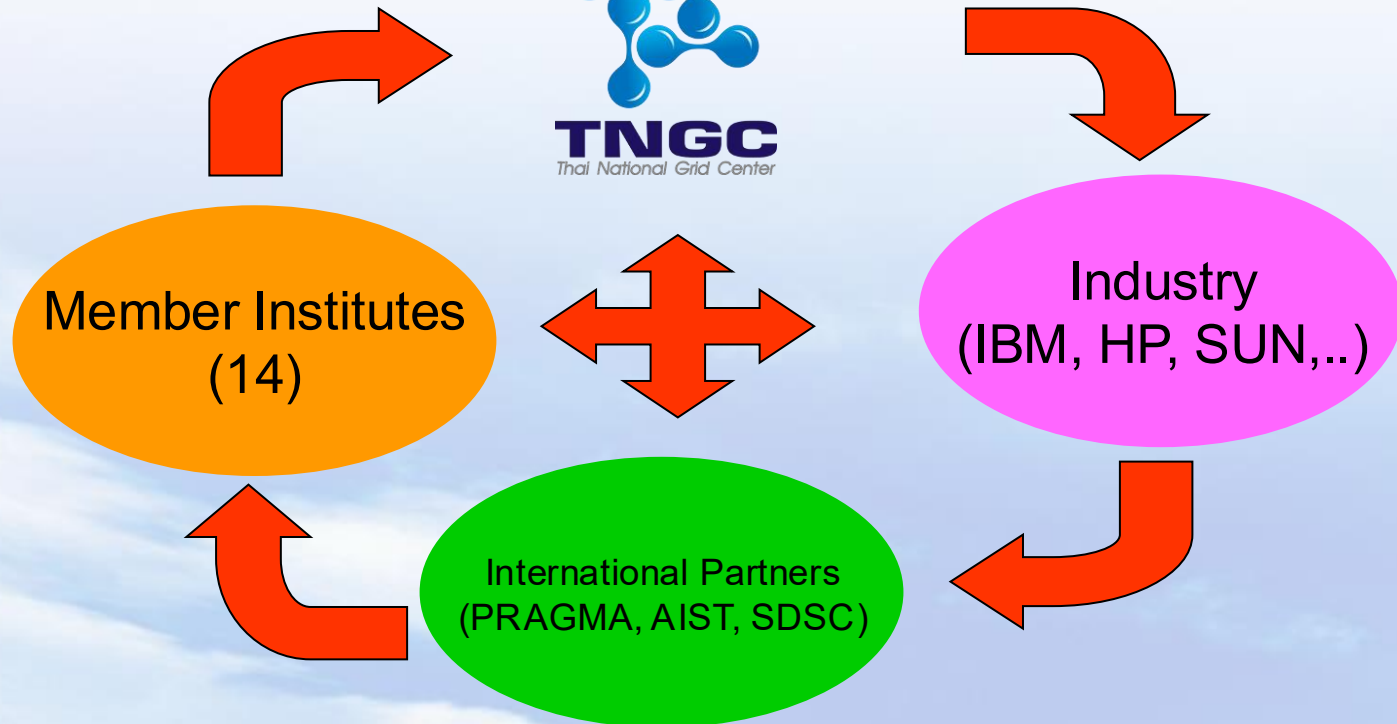
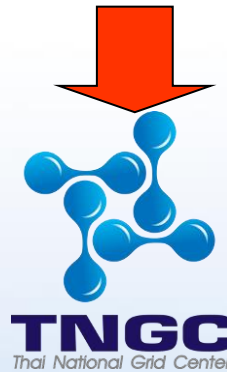
- **A National Project under Software Industry Promotion Agency (Public Organization) , Ministry of Information and Communication Technology**
- **The Thai National Grid Project is operated by the Thai National Grid Center**
- **GOAL**
 - **Building an e-infrastructure using grid as an enabling technology**



The Collaborations



Policy , Funding





Founding Members



Asian Institute of Technology (AIT)
Chaingmai University (CMU)
Chulalongkorn University (CU)
Kasetsart University (KU)
Khonkhan University (KKU)
King Mongkut's Institute of Technology North Bangkok (KMITNB)
King Mongkut's University of Technology Thonburi (KMUTT)
King Mongkut's Institute of Technology Ladkrabang (KMITL)
Mahidol University (MU)
Prince of Songkla University (PSU)
Suranaree University of Technology (SUT)
Silpakorn University (SU)
The Meteorological Department Of Thailand (TMD)
Walailak University (WU)



Thai National Grid Center (TNGC)

■ Vision

- Being an international excellence center in grid and HPC research

■ Missions

- Provision: Builds and Maintains a grid computing infrastructure in Thailand
- Innovation: Stimulates research and education in grid computing
- Promotion: Stimulates the adoption of grid technology in the industry, government, and education. Assists in setting up the national policies on grid computing
- Education: Creates knowledge man power in Grid Computing

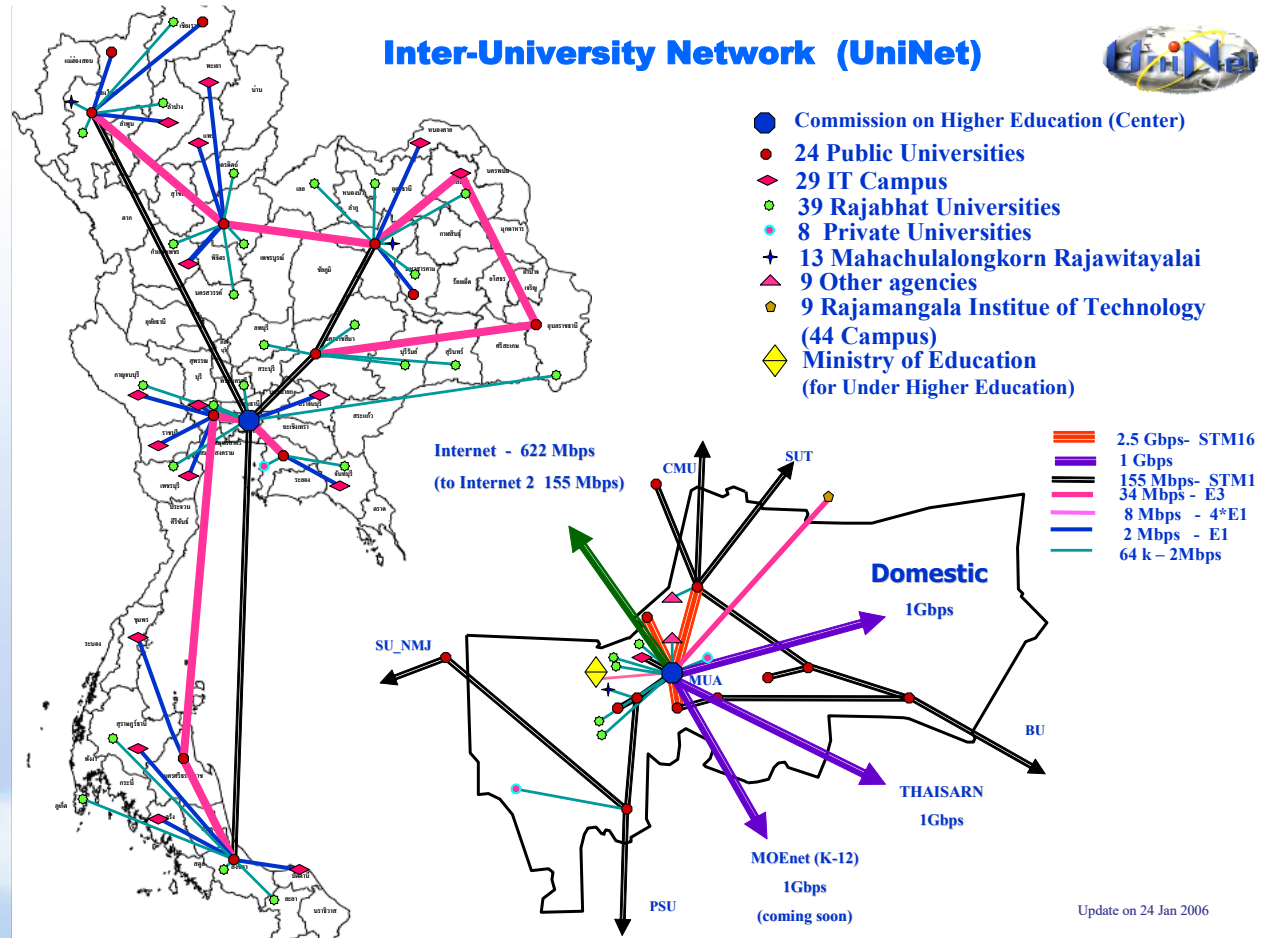




ThaiGrid
Thai National Grid Project

Connectivity

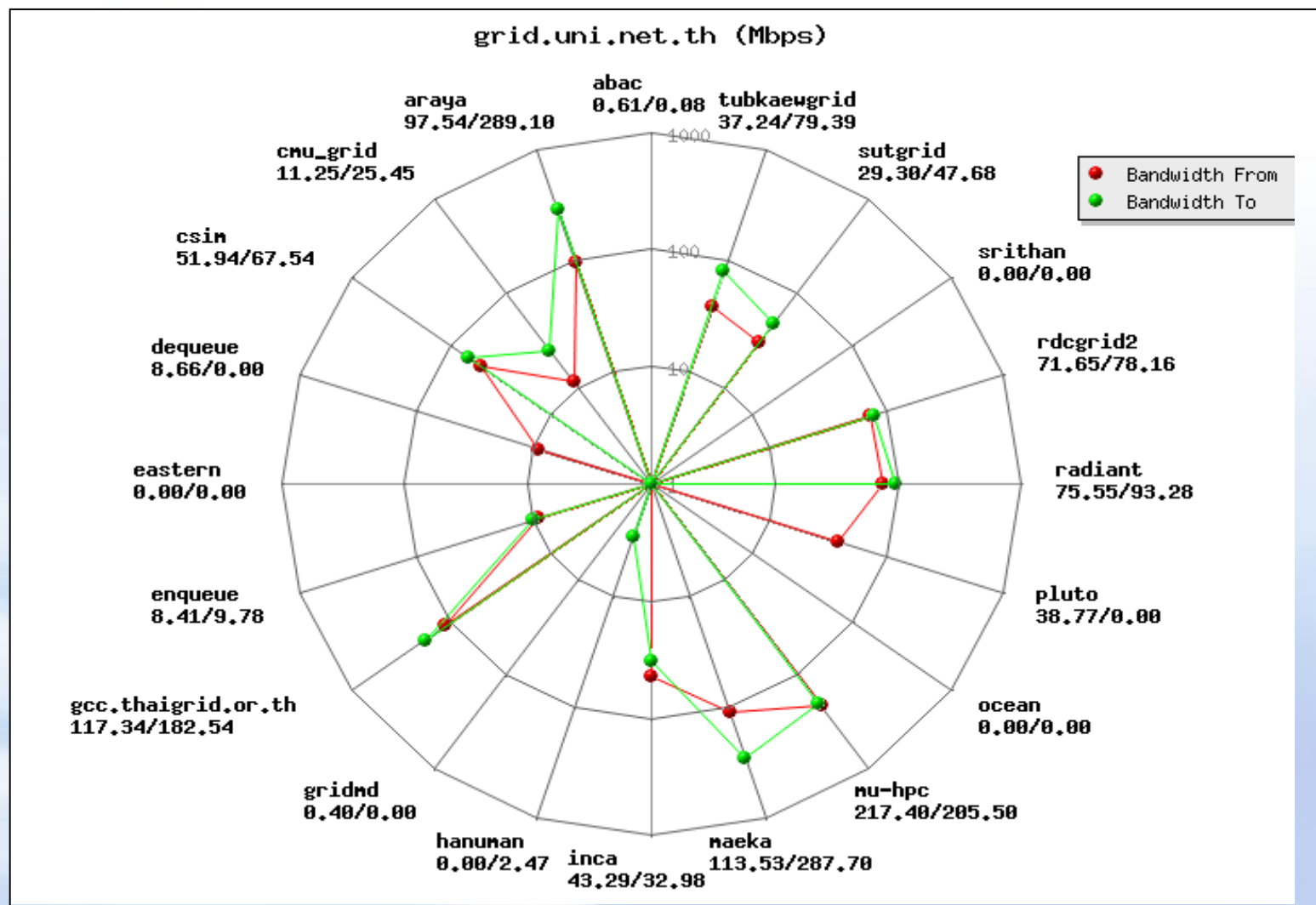
- Uninet
 - Large academic network
 - Gateway to Internet2 and Tien2
- Moving to Lambda network and 10Gbps speed





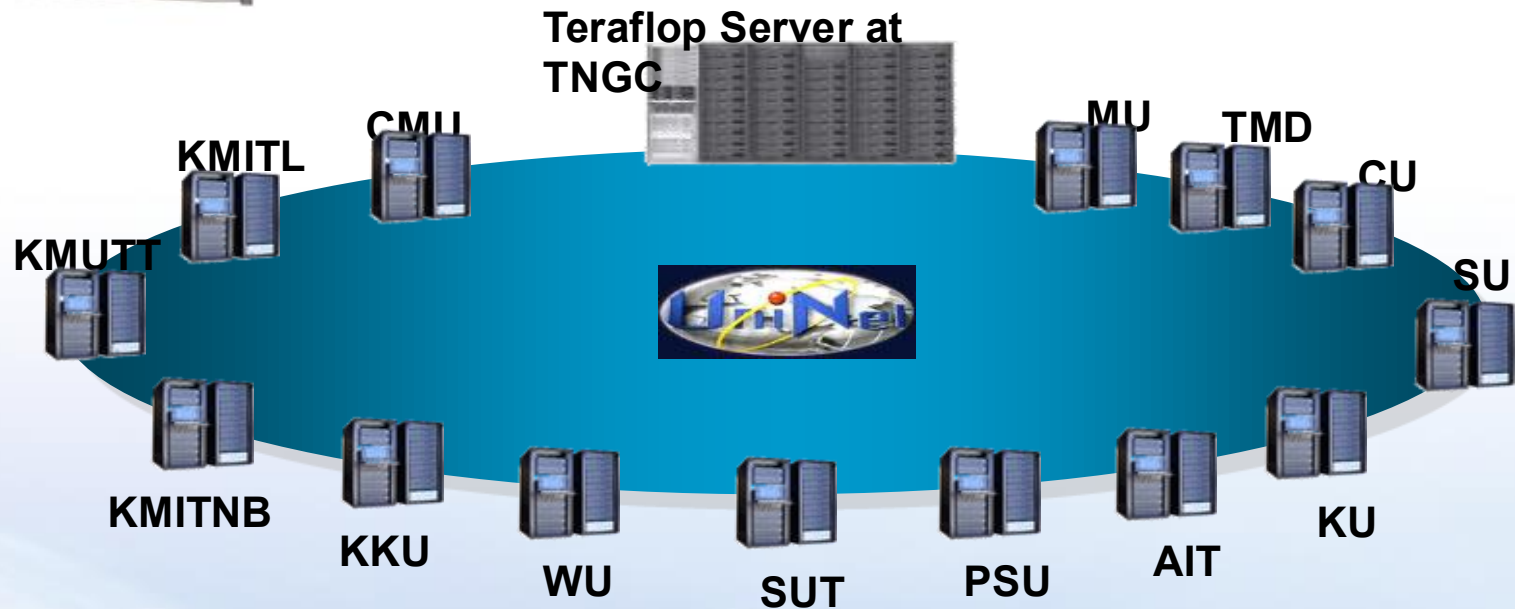
ThaiGrid
Thai National Grid Project

Current Network Speed





Computing Infrastructure



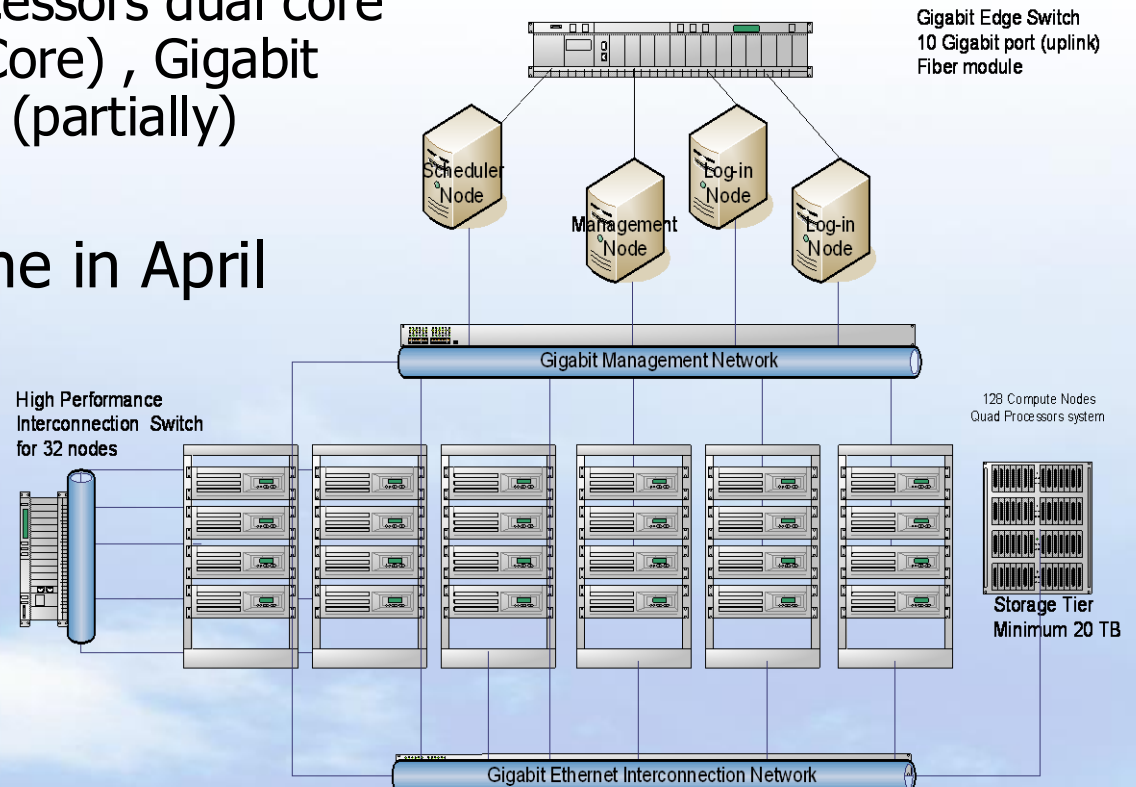
- First Phase: 80 dual processor nodes (160 CPU) has been installed in 14 founding institutes (IBM X series)
- TNGC has two IBM X3550 cluster (Araya, Sunyata) for POF application testing

ThaiGrid 2006



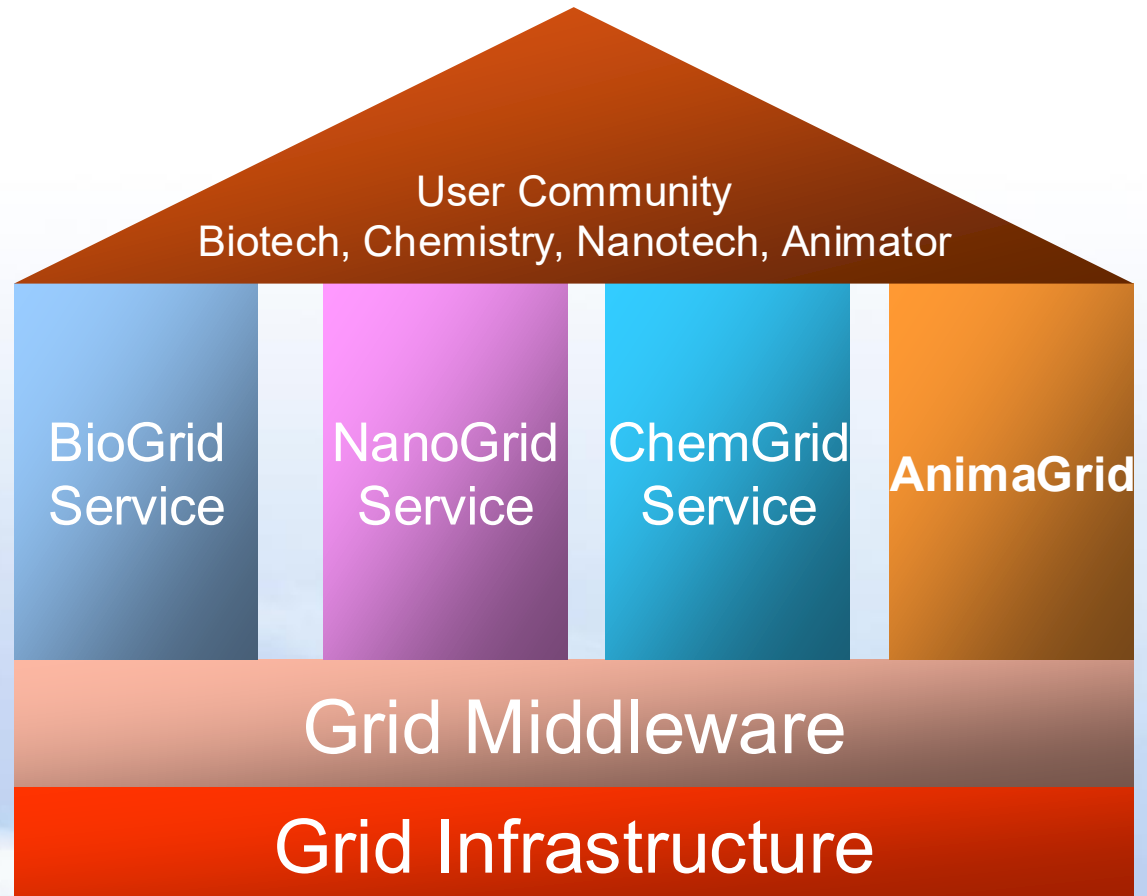
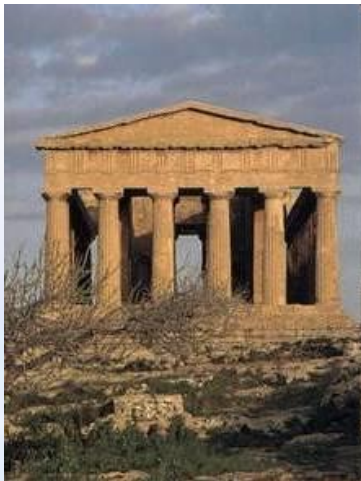
Teraflop Supercomputing System

- TNGC will also be equipped with a teraflop supercomputer
 - 200 nodes dual processors dual core XEON EMT64 (800 Core) , Gigabit Ethernet, Infiniband (partially)
 - HP DL360G5
- Expected to be online in April



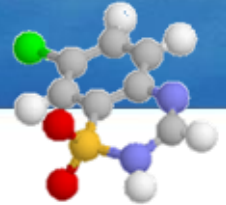


Making the Grid Useful

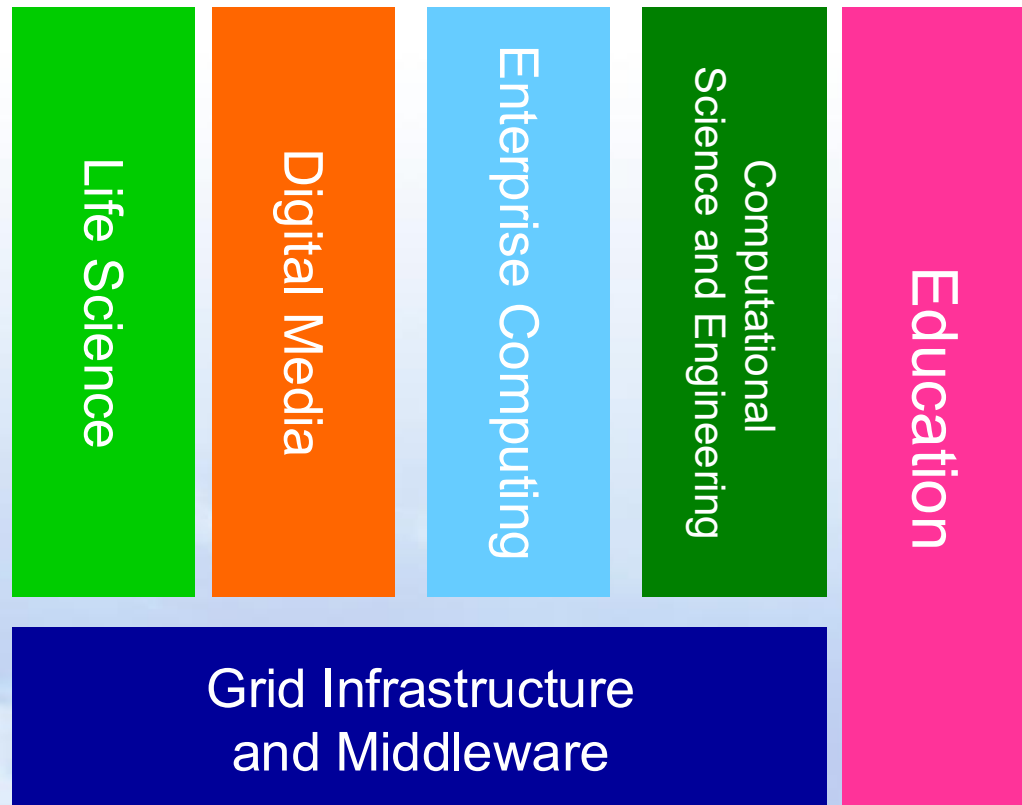


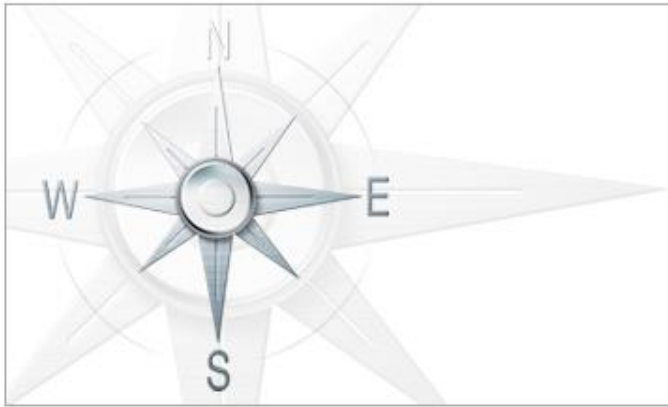


5 Initial Focus Areas

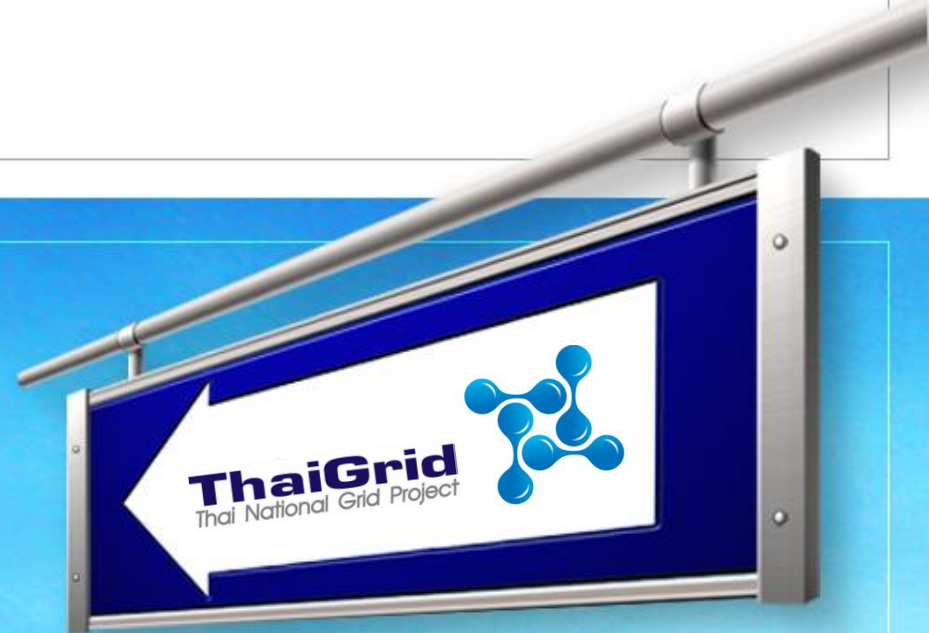


- **Life Science**
 - Drug Design
- **Computational Science and Engineering**
 - Geoinformatics, CFD
- **Digital media**
 - Rendering
- **Enterprise computing**
 - WS/SOA Enterprise computing application
 - Industrial Simulation and Modeling
 - Financial analysis
- **Education**
 - E-learning, collaborative environment
 - Grid education





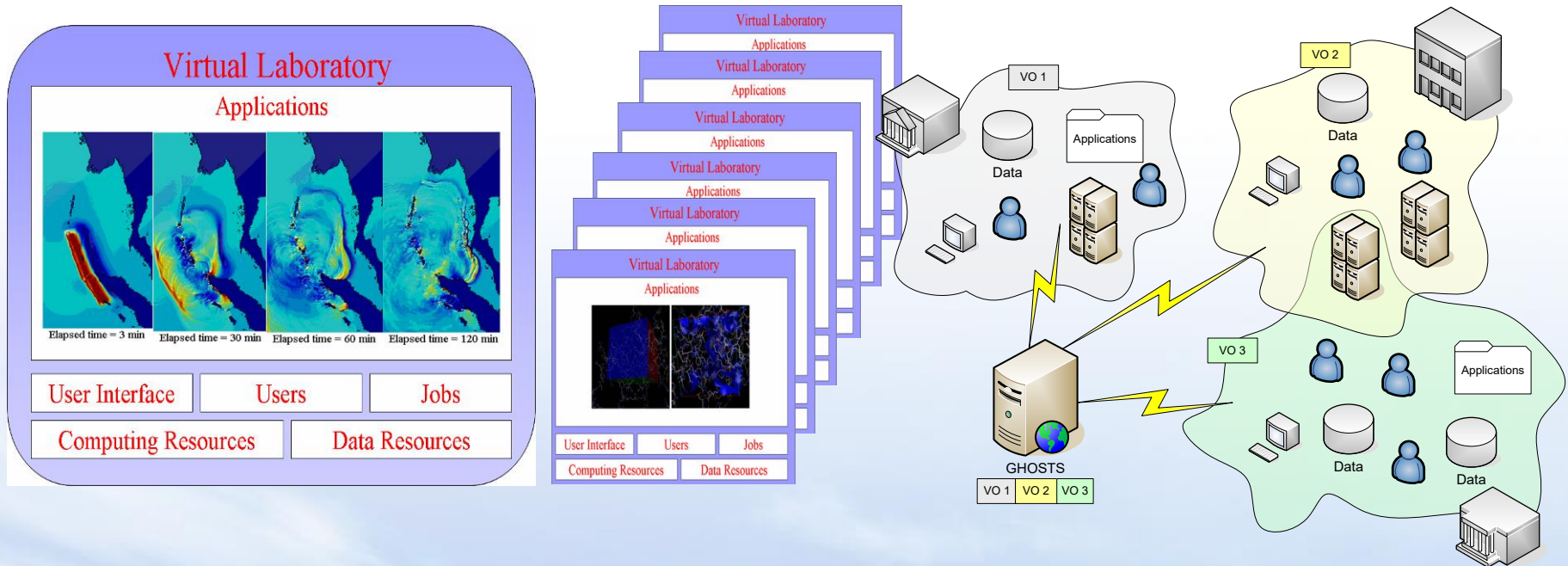
สำนักงานส่งเสริมอุตสาหกรรมซอฟต์แวร์แห่งชาติ (องค์การมหาชน)
Software Industry Promotion Agency (Public Organization)



Some of Our Current Research Projects



Virtual Research Center



- Explores the use of the Grid technology to build an advanced collaboration environment for researchers
 - Resources sharing
 - Collaborative environment over grid and advanced network
 - Funded by Thai National Grid Project

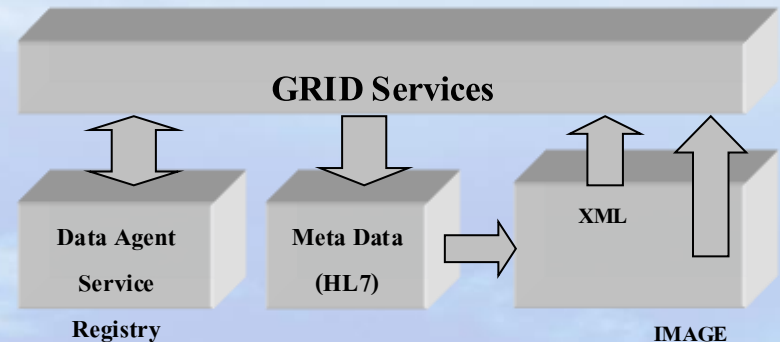
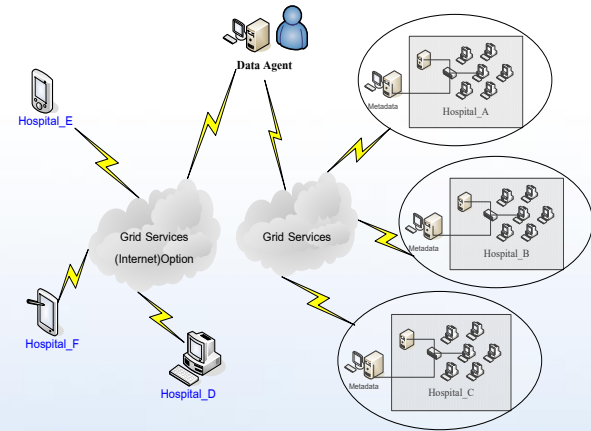


Medical Grid

Suwachai Siemoon, Somkiat Chatchuenyot, Dr. Panjai Tantasawong

Computer Science Department, Silpakorn University

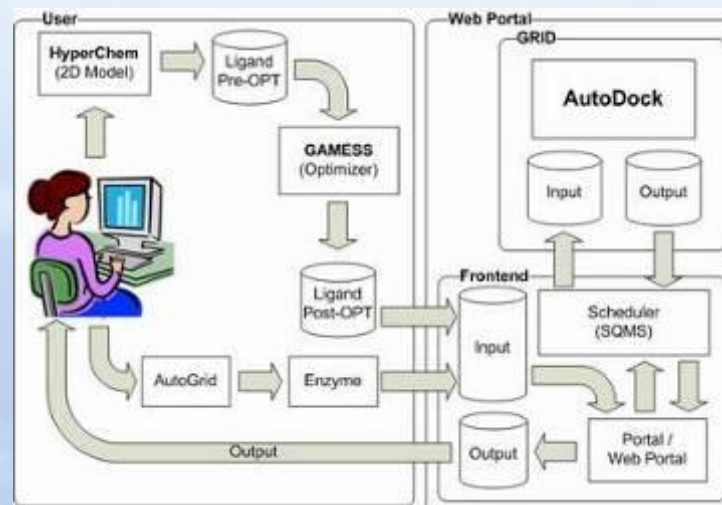
- To build a distributed information management system using DataGrid technology.
- Apply the distributed information management system to test on exchanging and managing the Patient Records System, which is record complexity and, different format and large size of data.



ThaiGrid Drug Discovery Infrastructure



- Partners:
 - Dr. Chak Saengma, KU
 - Somsak Sriprayoonsakul, TNGC
 - IBM Thailand
- Challenges
 - From over 10000 active compounds available from Thai medicinal plants database, find a smallest set of compounds that has a potential to be used as a drug
 - Very compute intensive. Several month of computing time
- Solution: Use grid to increase computing power to 10-100 times
 - Distributed screening process to hundred of hosts on the Grid
- Benefit
 - Speeding up time to results
 - Allow scientists to spend more time analyzing results





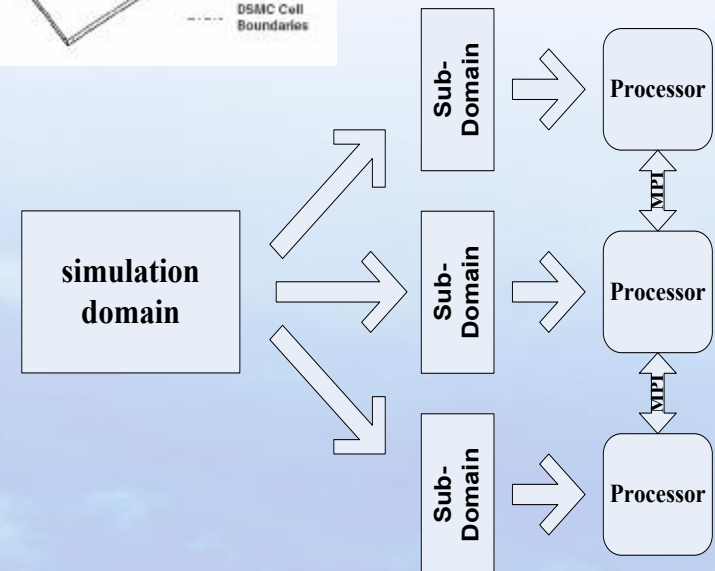
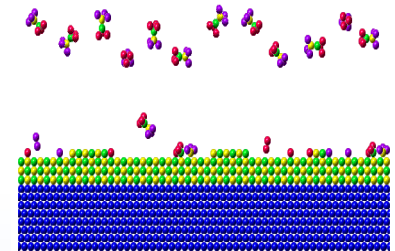
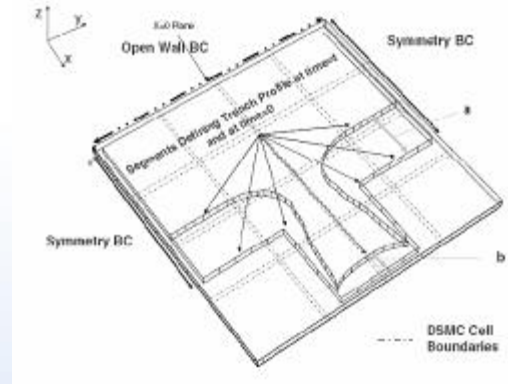
The Parallel Simulation Model for Thin Film Deposition Using the DSMC Method

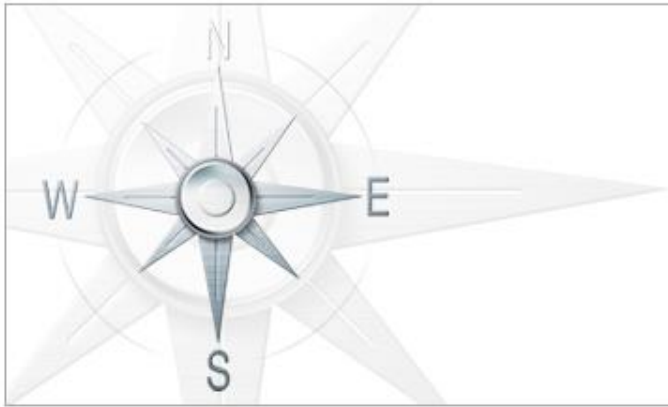
Dr. Tiranee Achalakul

Dr. Rardchawadee Silapunt

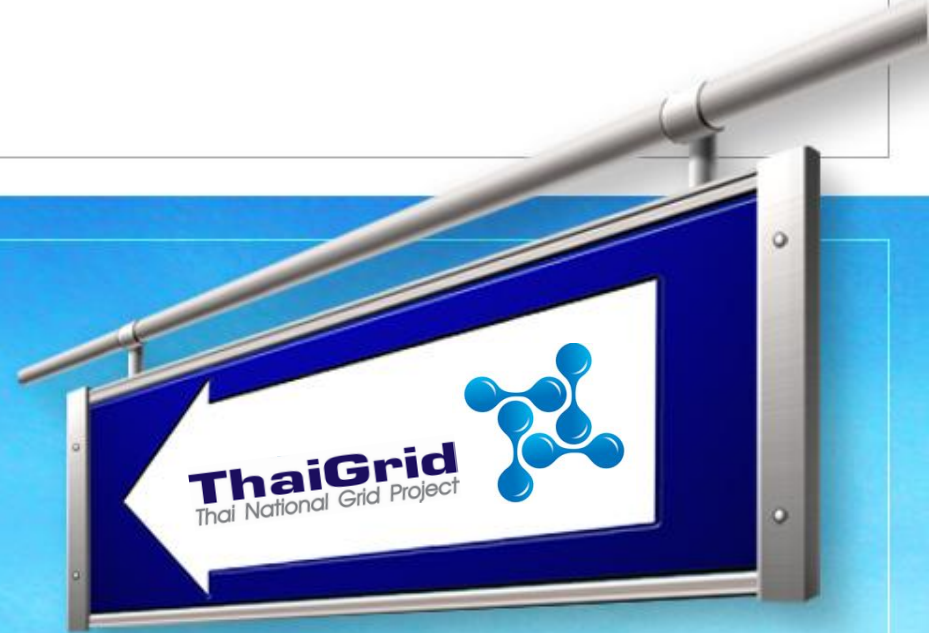
KMUTT, Thailand

- Thin film deposition process is used for many manufacturing industries including very large scale integrated circuits, optics, magnetic, hard and corrosion resistant coatings, micro-mechanics.
- Grid system will be used to accelerate the computation and simulation process
- Part of TNGP/ Hewlett Packard Join Research Program





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Software Industry Promotion Agency (Public Organization)



In Summary



Challenges

- The experience in building the ThaiGrid system presents several major challenges
 - **Firewall Configuration:** most universities lock out most non-standard network ports that are needed for the grid to function.
 - **Network Reliability and Bandwidth:** TNGC experienced a much lower sustainable bandwidth than expected throughout the network due to the incorrect configuration, the complex firewall rules, spam control, and the traffic shaping.
 - **Middleware configuration complexity:** Globus configuration can be complicated for novice system administration.



International Collaborations

- The experience also tells us that collaborations are much needed
 - PRAGMA: SCMS - the monitoring Technology
 - Tokyo Institute of Technology: Ninf-g Grid RPC
 - SDSC: Rocks and ThaiGrid Roll
 - OGF: GIN Testbed
 - University of Utah: CSE-Online
 - OMII-UK: MyGrid Project



Conclusions

- We hope to provide a strong e-infrastructure in order to accelerate the research and development in Thailand.
- We aim to promote the use of Grid technology in Thailand.
- We seek international collaborations at any level.



Joining the Ranks of the Technological Advancement in Grid Computing

<http://www.thaigrid.or.th>